

Raphael Bessin

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Seeking Co-op: January 2027 – August 2027 (4 - 8 months)

EDUCATION

Northeastern University, Khoury College of Computer Sciences

Boston, MA

Candidate for Bachelor of Science in Computer Science

Sep. 2025 – May 2029

Awards: GPA: 4.00/4.00 | Dean's List: Fall 2025 - Spring 2026 | John Martinson Honors College

Relevant Coursework: Object-Oriented Design, Algorithms & Data Structures, Mathematical Foundations of AI, Computer Systems

French American School of New York

New York, NY

US High School Diploma & International French Baccalaureate

Sep. 2021 – Jun. 2025

Awards: Ranked 1st in grade | Mention Très Bien (Highest Honors) | High Honors Math & Physics

SKILLS

Languages: Python, Java, JavaScript, TypeScript, HTML, CSS, SQL, GDScript, C#

Frameworks & Libraries: React, Next.js, Spring, Dash, Node.js, Express, NumPy, Vite, Pandas

Tools & Databases: PostgreSQL, Claude Code, Git, Docker, Swagger, MongoDB, Firebase, Oracle DB, Prisma, SQLAlchemy, Unity, Godot

Spoken Languages: English (native), French (native), German (intermediate)

Interests: Long-Distance Running, Hiking, Nature Photography, Neuroscience, Travel, Video Games

EXPERIENCE

FADEL

New York, NY

Software Engineering Intern

May 2026 – Jul. 2026

- Architected an API proxy unifying 3 backend systems across 13 domains for 300+ clients, with live testing and documentation.
- Integrated 63 API endpoints onto the proxy; shipped 20+ into FADEL's production backends including enterprise SSO (Google, Microsoft, SAML/OKTA) and an async Invoice Payment batch API with URL-based webhook callbacks and exponential backoff.
- Built Claude Code agents for database querying and bulk testing to surface silent errors and type truncations across upstream services, authoring 10+ remediation proposals; discovered and patched an SQL injection vulnerability in production.

Northeastern Electric Racing

Boston, MA

Software Lead, Parametric Simulation (NERSim)

May 2025 – Present

- Leading software development for NERSim, NER's vehicle dynamics simulator; building onboarding to bring on new developers.
- Replaced polling with WebSockets, cutting flickering by 80%+, and led a UI/UX overhaul including a React graph editor and presets.
- Engineered a CI/CD data-validation pipeline surfacing 8 broken data entries on a self-hosted GitHub Actions runner, performing automated schema verification against production snapshots with fail-safe deploy-time reserialization and error handling.
- Authored Claude Code harness, restricted .env access, and built a post-edit hook auto-running code quality tools.
- Architected a unit conversion system using invertible functions to support 20+ bidirectional conversions without duplicating logic.

Software Member, Project Management Dashboard (FinishLine)

Sep. 2025 – Apr. 2026

- Implemented global car filtering on a custom project management platform, replacing duplicated logic with a single reusable middleware-backed component across 20+ endpoints, giving 200+ users consistent, site-wide filtering for the first time.
- Proposed and built FinishLine's first developer documentation system for 30+ developers, a statically generated Docusaurus site auto-generating Claude Code skills from onboarding docs (DB seeding, API/hooks) to avoid having two separate sources of truth.

PROJECTS

CurbYourCarbon | JavaScript, Performance API, ElectricityMaps API | github.com/rbessin/CurbYourCarbon

2026

- Won Social Impact & Sustainability at HackBeanpot; built the initial MVP and led a 3-person team through the full feature set.
- Engineered a CO2 estimation engine modeling network and device-level emissions with regional-to-global carbon-intensity fallback, converting results into 3 real-world equivalencies (miles driven, phone charges, trees); published extension to Chrome Web Store.
- Designed a historical dashboard and gamified goal-tracking system with streaks and 8 achievements to drive behavior change.

Signal Relay | Godot, GDScript | github.com/rbessin/signal-relay

2025

- Designed a chip abstraction system for a digital logic simulator, enabling the composition of 7 base gate types and edge-triggered sequential logic into arbitrarily nested custom chips generated via automatic pin detection, with circuit and component persistence.
- Reduced multi-minute freezes from eager component instantiation to seconds via a lightweight/full two-state architecture.
- Developed an event-driven propagation system that recursively re-evaluates only downstream gates with changed outputs.

Stride Editor | Next.js, TypeScript, Quill.js | github.com/rbessin/stride-editor

2025

- Powered a browser-based text editor via a fourgram model with trigram/bigram/unigram fallback, trained on 10 Gutenberg books.
- Built a data pipeline to tokenize and prune n-gram models at build time, plus VSCode-style file navigation via File System Access API.